# Algebra II with Trigonometry Chapter 7.3

**Study Guide (Gemini)**

Saved tutoring response with MathJax rendering. Original PDF file: 4592201_alg-2trig-h-chp-7-3-note-docx.pdf

Here is a comprehensive study guide and summary based on the provided Algebra 2 Trig/Honors document.

### 1. Summary of the Document

This document is a guided notes packet for a Chapter 7.3 lesson on advanced trigonometric formulas. It introduces five major categories of trigonometric identities: Double-Angle, Half-Angle, Power-Lowering, Product-to-Sum, and Sum-to-Product formulas. Following the primary formulas, the packet provides 10 sets of example problems designed to teach students how to evaluate trigonometric functions, simplify expressions, find exact values of non-standard angles, integrate inverse trigonometry, and prove complex trigonometric identities.

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### 2. Core Definitions, Theorems, and Formulas

Below are the core formulas provided on the first page of the document:

#### I. Double-Angle Formulas

- $\sin(2x) = 2\sin x \cos x$
- $\cos(2x) = \cos^2 x - \sin^2 x$
- *Alternative 1:* $\cos(2x) = 1 - 2\sin^2 x$
- *Alternative 2:* $\cos(2x) = 2\cos^2 x - 1$
- $\tan(2x) = \frac{2\tan x}{1-\tan^2 x}$

**II. Half-Angle Formulas** *(Note: The $\pm$ is chosen based on the quadrant in which the angle $u/2$ lies).*

- $\sin\left(\frac{u}{2}\right) = \pm\sqrt{\frac{1-\cos u}{2}}$
- $\cos\left(\frac{u}{2}\right) = \pm\sqrt{\frac{1+\cos u}{2}}$
- $\tan\left(\frac{u}{2}\right) = \frac{1-\cos u}{\sin u} = \frac{\sin u}{1+\cos u}$

**III. Formulas for Lowering Powers**

- $\sin^2 x = \frac{1-\cos(2x)}{2}$
- $\cos^2 x = \frac{1+\cos(2x)}{2}$
- $\tan^2 x = \frac{1-\cos(2x)}{1+\cos(2x)}$

**IV. Product-to-Sum Formulas**

- $\sin u \cos v = \frac{1}{2}[\sin(u+v) + \sin(u-v)]$
- $\cos u \sin v = \frac{1}{2}[\sin(u+v) - \sin(u-v)]$
- $\cos u \cos v = \frac{1}{2}[\cos(u+v) + \cos(u-v)]$
- $\sin u \sin v = \frac{1}{2}[\cos(u-v) - \cos(u+v)]$

**V. Sum-to-Product Formulas**

- $\sin x + \sin y = 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$
- $\sin x - \sin y = 2\cos\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)$
- $\cos x + \cos y = 2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$
- $\cos x - \cos y = -2\sin\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)$

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## 3. Worked Study Guide

Since the provided document is a blank notes packet, let's walk through the *strategies* required to solve the types of examples presented.

**Skill A: Evaluating Double and Half-Angles using Quadrants (Examples 1, 5, 8)**

- *The Concept:* If you are given a ratio like $\tan x = -4/3$ in Quadrant II, you must find $\sin x$ and $\cos x$ before using the double/half angle formulas.

- *Step 1:* Draw a right triangle in the specified quadrant.
- *Step 2:* Use the Pythagorean theorem ($a^2 + b^2 = c^2$) to find the missing side (the hypotenuse is always positive).
- *Step 3:* Determine $\sin x$ and $\cos x$. In QII, sine is positive and cosine is negative.
- *Step 4:* Plug these values into your Double or Half-Angle formulas. *Tip: For Half-Angle formulas, divide the given domain by 2 to figure out which quadrant $x/2$ lives in so you know whether to use the $+$ or $-$ sign.*

**Skill B: Finding Exact Values of Non-Standard Angles (Example 3)**

- *The Concept:* You can find the exact value of angles like $22.5^\circ$ because it is exactly half of a standard unit circle angle ($45^\circ$).
- *Step 1:* Set $u/2 = 22.5^\circ$, which means $u = 45^\circ$.
- *Step 2:* Use the half-angle formula. Since $22.5^\circ$ is in Quadrant I, the result will be positive.
- *Step 3:* Evaluate $\sqrt{\frac{1-\cos(45^\circ)}{2}}$ using $\cos(45^\circ) = \frac{\sqrt{2}}{2}$ and simplify the complex fraction.

**Skill C: Lowering Powers (Example 2)**

- *The Concept:* You need to reduce expressions with exponents (like $\sin^4 x$) to expressions where cosine is only to the first power.
- *Step 1:* Rewrite the expression using squares: $\sin^4 x = (\sin^2 x)^2$.
- *Step 2:* Substitute the power-lowering formula: $\left(\frac{1-\cos 2x}{2}\right)^2$.
- *Step 3:* Expand the binomial (FOIL). You will end up with a $\cos^2(2x)$ term. Apply the power-lowering formula *again* to that specific term.

**Skill D: Proving Identities (Example 10)**

- *The Concept:* Show that the left side of an equation perfectly matches the right side.
- *Strategy:* Always start with the more complicated side. Look for structural clues. For example, if you see $\frac{1-\cos 2x}{\sin 2x}$ (Ex 10c), immediately replace the numerator and denominator with their double-angle equivalents (

$\cos 2x = 1 - 2\sin^2 x$ and $\sin 2x = 2\sin x \cos x$) and cancel terms until it matches the other side.

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**4. Practice Problems with Answers**

Here are five practice problems taken directly from the blank examples in the document, solved for you to check your understanding.

**Problem 1 (From Example 3a):** Use an appropriate Half-Angle Formula to find the exact value of $\cos\left(\frac{3\pi}{8}\right)$.

- *Answer:* Let $\frac{u}{2} = \frac{3\pi}{8}$, so $u = \frac{3\pi}{4}$.

Because $\frac{3\pi}{8}$ is in Quadrant I, cosine is positive.
$\cos\left(\frac{3\pi}{8}\right) = +\sqrt{\frac{1+\cos(3\pi/4)}{2}} = \sqrt{\frac{1+(-\sqrt{2}/2)}{2}} = \sqrt{\frac{2-\sqrt{2}}{4}} = \frac{\sqrt{2-\sqrt{2}}}{2}$

**Problem 2 (From Example 4a):** Simplify the expression: $2\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$
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- *Answer:* This perfectly matches the structure of the Double-Angle formula: $\sin(2x) = 2\sin x \cos x$, where $x = \frac{\theta}{2}$.

Therefore, the expression simplifies to $\sin\left(2 \cdot \frac{\theta}{2}\right) = \sin(\theta)$.

**Problem 3 (From Example 9a):** Find the sum for the product $\sin(2x)\sin(3x)$.

- *Answer:* Use the Product-to-Sum formula: $\sin u \sin v = \frac{1}{2}[\cos(u-v) - \cos(u+v)]$. Let $u = 2x$ and $v = 3x$.

$\frac{1}{2}[\cos(2x-3x) - \cos(2x+3x)]$ $\frac{1}{2}[\cos(-x) - \cos(5x)]$ *Note: Because cosine is an even function,* $\cos(-x) = \cos(x)$. Final Answer: $\frac{1}{2}\cos x - \frac{1}{2}\cos(5x)$.

**Problem 4 (From Example 10a):** Prove the identity: $\sin(8x) = 2\sin(4x)\cos(4x)$.

- *Answer:* Start with the right side, which resembles the Double-Angle formula for sine: $\sin(2\theta) = 2\sin\theta\cos\theta$.

Let $\theta = 4x$. Substitute $\theta$ into the formula: $2\sin(4x)\cos(4x) = \sin(2(4x))$. Simplify the argument: $\sin(8x)$. The left side equals the right side.

**Problem 5 (From Example 2):** Use formulas for lowering powers to rewrite $\sin^4 x$ in terms of the first power of cosine.

- *Answer:*

1. Rewrite: $(\sin^2 x)^2$ 2. Substitute: $\left(\frac{1-\cos 2x}{2}\right)^2$ 3. Expand: $\frac{1-2\cos 2x+\cos^2 2x}{4}$ 4. Lower the remaining squared term using $\cos^2(2x) = \frac{1+\cos 4x}{2}$: $\frac{1-2\cos 2x+\left(\frac{1+\cos 4x}{2}\right)}{4}$ 5. Multiply the numerator and denominator by 2 to clear the complex fraction: $\frac{2-4\cos 2x+1+\cos 4x}{8} = \frac{3-4\cos 2x+\cos 4x}{8}$

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**5. Assumptions or Ambiguities Resolved**

- **Blank Examples:** The provided document is a teacher's guided notes template, meaning Examples 1 through 10 contained prompts but no solutions. To fulfill the prompt's request for a "worked study guide" and "practice problems," I assumed the role of the tutor and generated the correct mathematical steps and solutions for those specific blank examples.
- **Half-Angle Plus/Minus ($\pm$) Sign:** The formulas on page 1 list $\pm$ for the half-angle formulas for sine and cosine but do not explicitly state *how* to choose the sign. I assumed standard trigonometric curriculum rules, clarifying in the study guide that the sign is determined by checking which quadrant the target angle ($u/2$) resides in.
- **OCR Formatting Variations:** The OCR text generation occasionally struggled with mathematical formatting (e.g., rendering $\sin^4 x$ poorly in Example 2). I relied exclusively on the visual screenshots of the PDF to ensure all formulas, variables, and exponents were transcribed properly in standard math notation (LaTeX).